

In every organization, gender stereotypes reduce organizational efficiency and waste productive energy: a systems thinking perspective

Piero Mella

Department of Economics and Management, University of Pavia, Pavia, Italy

Abstract

Purpose – Stereotypes are simplified and widely shared visions held by a social group regarding a place, object, event or recognizable set of people united by certain characteristics or qualities. They are “dangerous” mental models because they are widely disseminated, devious and capable of acting even unconsciously in individuals, social groups and organizations altering the rationality of assessments and choices and producing discrimination and prejudice. Stereotypes acritically extend from a characteristic of a significant percentage of a category to the totality of individuals. The process of generalization triggered by a stereotype produces the error of discrimination and prejudice. There are numerous forms of stereotypes, but this study takes into account gender stereotypes because they act pervasively, often subtly, to reduce “productivity”. People who are aware of being discriminated perceive an unsatisfactory fulfillment of their motivations, which reduces their incentive to improve their performance. Since productivity measures the efficient use of energy from working in production processes, the author believes that wherever gender stereotypes are at play, there is a productive “waste of energy”, an inefficiency in work activity with harmful effects for organizations of all kinds, including families.

Design/methodology/approach – The work aims to demonstrate that wherever gender stereotypes are at play, a “waste of energy” manifests itself in terms of productivity, representing an inefficiency in work activity with harmful effects for organizations of all kinds, including families. To describe the negative effects stereotypes produce in organizations, some models are presented based on the methods and language of systems thinking. These models, although typically qualitative, are capable of exploring the most accepted theories in the literature: tournament theory, the Pygmalion effect, the Galatea effect, self-fulfilling prophecies, the Queen bee syndrome, the role congruency theory, the glass ceiling theory (“think manager, think male” and “family responsibilities wall”). The paper follows a predominantly organizational and corporate approach, although the copious literature on stereotypes belongs largely to the area of social psychology and organization studies.

Findings – The paper does not consider the psychological origin of stereotypes but highlights their use as routines-shortcuts for evaluations and decisions demonstrating that, when adopted in social systems and within organisations, stereotypes produce different forms of discrimination: in social rights, in work, in careers and in access to levels of education and public services, reducing performance and limit potential. The paper also examines some ways gender and culture stereotypes can be opposed, presenting a change management strategy and some concrete solutions proposed by the process–structure–culture model for social change (PSC model).

Research limitations/implications – The main limitation of the work is that it focuses on gender stereotypes, choosing not to consider the “intersection effect” of these with other stereotypes: racial stereotypes, religious stereotypes, color stereotypes, age stereotypes, sex and sexual orientation stereotypes, and many others, whose joint action can cause serious inefficiencies in organizational work.

Practical implications – As stereotypes are a component of social culture and are handed down, by use and example, from generation to generation, the maintenance over time of stereotypes used by individuals to evaluate, judge and act can be seen as an effect of the typical action of a combinatory system of diffusion, which can operate for a long time if not effectively opposed. If PSC model indicates the strategy for carrying out this opposition.



Social implications – With regard to gender stereotypes, it should be emphasized that in organizations and social systems, “gender diversity” should be considered an opportunity and not as a discriminating factor and thus encouraged by avoiding harmful discrimination. In fact, this diversity, precisely because of the distinctive characteristics individuals possess regardless of gender, can benefit the organization and lead to an increase in organizational and social performance. The United Nations 2030 Agenda for Sustainable Development (2020) Goal 5: Achieving gender equality and empowering all women and girls is examined in this context.

Originality/value – This study views the action of gender stereotypes as especially harmful “mental models”, highlighting the distortions they cause in the allocation of productive energy in society, groups and organizations. The paper follows a predominantly organizational and corporate approach, although the copious literature on stereotypes belongs largely to the area of social psychology. Using the “logic” and “language” of systems thinking, theories and models that describe and interpret the distorting effects of organizational choices based on stereotypes rather than rational analysis are highlighted. The action of stereotypes and their persistence over time can also be described using combinatory systems theory. With this paper, the author hopes that by acting on the three wheels of change highlighted by the PSC model, through legal provisions, control tools and actions on the culture operated by educational and social aggregative institutions, it should not be impossible to change the prevailing culture so that it becomes aware of the harmful influence of gender stereotypes and other discriminatory mental models and come to reject them. The author hopes this paper will help to understand the need to make this change.

Keywords Mental models, Stereotype, Gender stereotypes, Tournament theory, Self-fulfilling prophecies, Congruency theory, The glass ceiling theory, Drivers of productivity, PSC (process–structure–culture) theory

Paper type Conceptual paper

1. Introduction – stereotypes are harmful mental models

In his excellent book *The Fifth Discipline: The Art and Practice of the Learning Organization* (1990, First Edition), Peter Senge defines *mental models* as “schemes” and “theories” that guide an individual in the decisions and choices he/she makes through his/her actions. Mental models have a profound influence on the way people act in two related ways

- (1) On the one hand, they are representations of the world that offer “subjective information” for human action; since a man normally observes things subjectively and selectively, two individuals with different mental models can observe the same phenomenon/event/object and describe it differently because they have looked at different sizes and dimensional states (Mella, 2020). Jay Forrester, the father of systems dynamics, explicitly clarifies that mental models are private images the individual builds by selecting concepts and relationships deemed useful for representing reality (Forrester, 1971, p. 213).

Even more clearly, in his book titled *Mental Models: Towards a Cognitive Science of Language, Inference and Consciousness* (1983), the psychologist Philip Johnson-Laird highlighted the cognitive function performed by mental models in the representation and understanding of the world

The psychological core of understanding, I shall assume, consists of having a “working model” of the phenomenon in your mind. If you understand inflation, a mathematical proof, the way a computer works, DNA, divorce, then you have a mental representation that serves as a model of an entity in much the same way as, say, a clock functions as a model of the Earth’s rotation. [. . .] Many of the models in people’s minds are little more than high-grade simulations, but they are none the less useful provided that the picture is accurate (Johnson-Laird, 1983, pp. 2, 4).

- (2) On the other hand, they are *also bearers of judgments* about the world; for this reason, mental models are “active” because they affect the *assessments* and *decisions* that underlie *action*; thus, they determine not only how individuals give meaning to the world but also how they act in it.

A mental model, therefore, guides the individual in his or her representations of the world while at the same time influencing the decisions and choices he or she makes when acting, especially in organizations. Peter Senge himself emphasizes the dual function of mental models in the formation of knowledge and in influencing our judgment, thereby guiding our decisions.

They influence our *understanding* of the world and how we take action, in which

Mental models are deeply ingrained assumptions, generalizations, or even pictures or images that influence how we understand the world and how we take action, in that we are not consciously aware of them and the effects they have on our behavior (Senge, 1990, p. 8).

However, they also influence our judgments since they are “active” and shape how we act. Their power in affecting what we do stems in part from their ability to affect what we see. “Two people with different mental models can observe the same event and describe it differently, because they’ve looked at different details” (Senge, 1990, p. 160).

It is difficult to grasp the mental models adopted by individuals since, being deeply rooted in the human subconscious, they form a large part of “tacit knowledge” to such an extent that they unconsciously influence people’s behavior (Spender, 1993; Polanyi, 2015). For this reason, when subjects with different mental models interact, not only can misunderstandings occur but violence and conflict as well.

Mental models are “conservative”: if the results obtained by applying the model are positive, the “schema” is strengthened; otherwise, it is “set aside,” and another behavioral model is constructed (Leonard and Sensiper, 1998). “Winning” models tend to remain subjective over time and spread within more or less broad social groups, to the point of becoming part of the shared *culture* of a society, understood in the broadest sense (Schein, 1990, p. 7). The natural *resistance to change* and “innate conservatism” arise precisely from the tendency to “preserve” mental models, and it takes commitment and effort to distance ourselves from a particular model.

Mental models are widespread even in organizations. At this level, the mental models involved are those shared within the organization, and they typically consist of operational, selective and evaluative procedures, which represent widely accepted organizational models and standardized decision-making rules, especially in the form of *organizational routines* (Nelson and Winter, 1982). Among the organizational mental models, we can also include policies, that is, implicit and explicit rules through which information is transformed into company decisions and actions.

Mental models, whether “right” or “wrong”, create a problem for the *organization* when they are tacit, i.e. when they exist below the level of awareness or are not made explicit by individuals. Since individuals are “obliged” to interact by “organizational constraints” in organizations, mental models – especially those that act below the conscious level – can constitute an obstacle to collaboration and organizational learning. Organizations therefore need a “discipline” that *teaches* and *obliges* individuals to recognize the subconscious action of mental models, encouraging (or forcing) individuals to explain and question them in order to verify their validity, share them or even modify them. This discipline – which Senge defines as the “Second discipline or the discipline of mental models” – leads individuals, groups and organizations themselves to increase their learning skills in order to improve their decision-making.

Following Senge, the “mental models discipline”, together with systems thinking, the fifth of five disciplines, is fundamental in constructing a “learning organization” and developing organizational learning, since it not only increases the group’s or the individual’s capacity to form a stock of shared knowledge but also facilitates the process for recognizing and modifying the group mental models in order to collectively decide in an effective way, as if the decision came from a single individual (Senge, 2006, p. 8).

Systems thinking (Mella, 2008, 2012) contributes to the mental models discipline as it allows us not only to make more evident the cause and effect links between the decisions and the results obtained based on the models but also to identify the strengthening loops that fortify those models and the balancing loops that hinder their change. To improve the efficiency of its organization, Shell – which was in great difficulty in the 1970s – adopted a new approach: the institutionalization of mental models. Arie De Geus, Shell's strategic coordinator, felt that long-term success depended on the speed of the process by which executives changed their mental models by sharing them with the company and the markets.

So the real purpose of effective planning is not to make plans but to change the microcosm, the mental models that these decision makers carry in their heads. And this is what we at Shell and others elsewhere try to do (Geus, 1988, p. 71). [...] We understand that the only competitive advantage the company of the future will have is its managers' ability to learn faster than their competitors. So the companies that succeed will be those that continually nudge their managers towards revising their views of the world. The challenges for the planner are considerable. So are the rewards (ibidem, p. 74).

Precisely because mental models are fundamental to our existence, some models have spread to become part of the *culture* of a given social group, population or organization; they are handed down from generation to generation and often strengthened, extending to an increasing number of individuals and becoming indispensable in decisions and actions. Many of these mental models are "harmful" to individuals and the *organizations* to which they belong because they distort the rationality of choices and produce undesired effects. This study will consider particular mental models – *stereotypes* – which, despite appearing perfectly rational for the individuals who use them, produce the harmful effects of reducing efficiency and performance, especially in organizations, and of generating and encouraging *discrimination*. The *stereotype models* are "harmful" to individuals and the *organizations* to which they belong because they *distort the rationality of choices* and produce undesired effects. The paper does not consider the psychological origin of stereotypes but highlights their use as routines-shortcuts for evaluations and decisions demonstrating that, when adopted in social systems and within organizations, stereotypes produce different forms of discrimination: in social rights, in work, in careers and in access to levels of education and public services, reducing performance and limit potential.

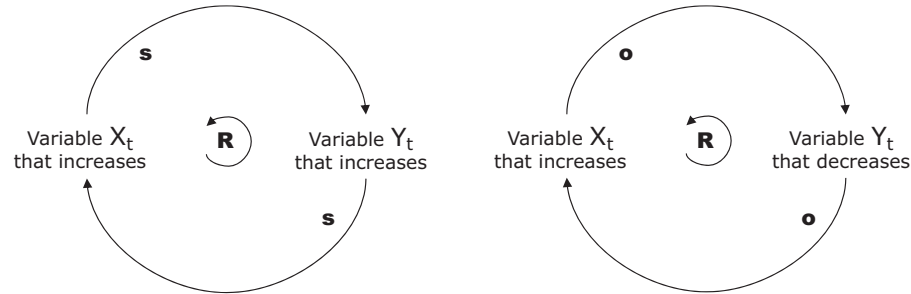
To address these issues, the paper is structured as follows: Section two recalls the origin of the term "stereotype" and indicates, without analyzing them, some of the most widespread (and harmful) stereotypes that generate discrimination and prejudice. Section three highlights the "mechanism" that generates the dissemination and maintenance of stereotypes within a community or organization, proposing an interpretation based on the typical action of a *combinatory system of diffusion*. Sections four and five focus on the distortive mechanisms that gender stereotypes induce in organizations, and some significant interpretative models of the harmful effects produced by such mental models in the work environment are examined with reference to hierarchical structures in which career advancement can be greatly influenced by gender discrimination that distorts evaluations based on merit and on the potential of the employee. Section six examines how gender stereotypes can adversely affect "productivity" by leading to a "waste of working energy." After recalling the factors on which business productivity depends, the paper demonstrates how those mental models can be a motivating factor, one of the most relevant elements on which productivity depends. Is it possible to counteract and control the "culture of stereotypes" to bring rationality back into organizational behavior? Since stereotypes are part of the culture of certain collectivities, actions must be taken to change this distorted culture, and section seven presents the solution offered by process-structure-culture (PSC) theory. In the construction of models, reference is made to the logic and language of systems thinking in the popular version proposed by Senge (1990; Mella, 2012), as indicated in Section 2.

2. Methodology – the role of causal loop diagrams

This paper does not represent quantitative empirical research, based on a statistical analysis of online data or questionnaires provided by the author; it is, instead, a conceptual paper primarily based on theoretical considerations, frameworks and models observed and represented according to the methodology and language of systems thinking, as formalized by [Senge \(1990\)](#) and [Mella \(2012\)](#). Modelling is an essential part of scientific activity; in my opinion, systems thinking is not a technique but an efficient methodology for building models about knowledge of a dynamic world. It is a qualitative methodology that, along with others, can be used in research, as is well evidenced in the works of James [Mahoney \(2007\)](#) and [Koivu and Damman \(2015\)](#).

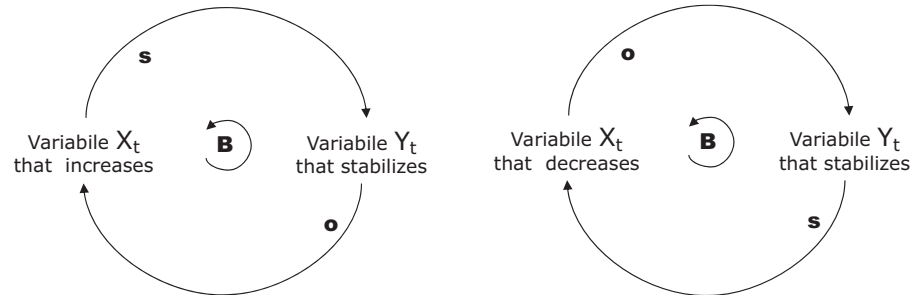
Systems thinking methodology assumes knowledge of the language of causal loop diagrams, which are fundamental models that represent not only *linear relationships* among variables forming “causal chains” but also, and above all, the *circular relationships* among interacting variables forming “loops” representing “circular links.” It may be helpful to remember that there are only two basic types of loops

- (1) *Reinforcing loops* [R] (shown in Model A) which, in successive repetitions of the system’s cycle, produce a reciprocal increase or reduction in the values of the two variables, which have an identical direction of variation: “s and s” or “o and o”, as shown below:



Model A. Reinforcing loops

- (2) *Balancing loops* [B] (Model B), which, with oscillations, maintain relatively stable the values of the connected variables.



Model B. Balancing loops

A causal loop diagram (Sterman, 2000) is, therefore, the model of a system of loops in which all the variables are linked together. The research design, which follows the systemic approach, predominantly employs causal loop diagram for translating the mental models incorporating the gender stereotype into formal systemic models to make some concepts easier and more accurate to understand, define and visualize.

The modeling must, of course, refer to existing and usually commonly accepted knowledge. The analysis of the literature was particularly focused on research concerning the effects gender stereotypes produce in organizational contexts, with particular reference to the negative effects of the stereotype on productivity when it conditions the selection and career paths of managers and employees in general, thereby affecting motivation and self-esteem as well as the willingness to remain in the organization. My research develops formal models and therefore attempts to represent and connect in a broad framework the multiple theories and aspects produced by the action of the gender stereotype. Following Huberman and Miles (1994), according to which a conceptual framework “lays out the key factors, constructs or variables and presumes relationships among them” (p. 440) and with the aim of expanding the interpretation beyond existing qualitative studies from the same discipline (Paterson *et al.*, 2009), the paper presents a construct in which each concept is examined and modeled distinctly, but within a broader system in which each concept is integrated, thereby allowing the reader to form a mental model of his own and possibly to develop it further. The main contribution of this line of research is, on the one hand, to explain the important relation between the gender stereotype and employee engagement, to improve organizational productivity and, on the other hand, to propose the logic of the PSC model as a method to address the gender stereotype problem.

3. The frame of reference – stereotypes: notion and typology. Discrimination and prejudice

Mental models that are referred to as “stereotypes” can greatly influence the behavior of people who adopt or are subject to them; their spread and maintenance over time within social groups and *organizations* is a relevant phenomenon. A brief discussion of the underlying meaning of the term “stereotype” is useful, as well as a mention of the large number and varieties of classes of stereotype which represent an integral part of our individual and collective culture.

The term “stereotype” was introduced into social systems by the journalist Walter Lippmann in his book *Public Opinion* (1922), in which, adopting a typographical term indicating the printing technique whereby a page is impressed and exactly reproduced in a single block of lead, he attributed to the term the meaning of rigid mental images, mental simplifications: that is, a particular well-established mental model both widely used and difficult to remove. Lippmann used the concept of stereotype to provide a clear and effective explanation of the process by which citizens’ opinions – when these reveal a clear mental model and are widespread – become public opinion, thereby generating inertia in the analysis of facts and often leading to preconceptions. He also analyzed the role of the mass media in the development of this process.

Real space, real time, real numbers, real connections, real weights are lost. The perspective and the dimensions of action are clipped and frozen in the stereotype (Lippmann, 1922, p. 100). What will be accepted as true, as realistic, as good, as evil, as desirable, is not eternally fixed. These are fixed by stereotypes, acquired from earlier experiences and carried over into judgment of later ones (*ibidem*, p. 107). The systems of stereotypes may be the core of our personal tradition, the defenses of our position in society. They are an ordered, more or less consistent picture of the world, to which our habits, our tastes, our capacities, our comforts and our hopes have adjusted themselves (*ibidem*, p. 63).

Stereotypes are usually defined as *simplified* visions (mental models) that are widely shared within a community or organization: those who follow a stereotype accept the opinion – without any *direct specific verification* – that a subject must have certain characteristics only because he belongs to a group that, on average, possesses them (Valian, 1998; Steele, 1997; Smith *et al.*, 2006;

Marx and Ko, 2012; Rosenthal and Overstreet, 2015). Generally speaking, a stereotype is formed in a social system when a characteristic of a not insignificant percentage of a category is extended to the totality of individuals. Therefore, at the root of a stereotype is a *process of generalization* that produces the error of *discrimination*; a stereotype is a simplified and widely shared vision about a place, object, event or recognizable group of people united by certain characteristics or qualities and an abstract and schematic mental model that can have a neutral, positive or negative meaning, which reflects the opinion of a social group with respect to other groups, producing harmful and irrational effects. Sometimes the stereotype is a caricature or reversal of some positive characteristics possessed by the members of a group, exaggerated to the point of becoming detestable or ridiculous, prompting people to *discriminate* and *reject* those who possess those characteristics.

Stereotypes are a pervasive feature of the cognitive process of individuals (Fiske *et al.*, 2002; Brown, 1986), distorting the use of information (Hudak, 1993) and producing systematic errors in its use. Following a stereotype in making evaluations and decisions is apparently “advantageous” since the model eliminates the effort of processing a personal judgment; however, at the same time, they hinder people from gaining knowledge of reality and lead to errors of judgment because they *push people to attribute a characteristic shared by the average member of a group even to those belonging to this group who do not possess it*, leading to possible distortions in coexistence, selection and actions. The stereotype, therefore, not only can help make sense of the world but often represents a “shortcut to action”, a kind of convenient “decision-making routine” that avoids the laborious evaluation process and produces in individuals a “seemingly coherent” worldview, capable of making them feel they are on the right side even without making an effort to verify their opinions (Kahneman, 2011, with comments by O’Brien, 2012). That is why stereotypes are so numerous, widespread, enduring and tenacious in society (Zawisza, 2018, online): their self-perpetuating quality is sustained by cognitive distortion (Hentschel *et al.*, 2019).

Stereotypes do not always concern individuals but can also refer to objects or animals (for example, spilling salt on the floor brings bad luck; if a black cat crosses your path, turn back), historical places or eras (for example, in medieval Florence people did not take care of their personal hygiene); they can be positive (for example, Italians love the opera; the French are romantic), negative (for example, Italians are all in the Mafia; all rock musicians take drugs) or neutral (for example, Christmas is not Christmas without snow and a crackling fire).

There is a significant number of stereotypes, which vary according to the areas over which social systems extend and in relation to the historically formed culture. Leaving it to the reader to find specific evidence and examples of this, among the most common are (SPEEXX, online; Canadian Human Rights Act, online) as follows

- (1) Gender stereotypes: these are generalizations about what men and women are like (Hentschel *et al.*, 2019); they reflect normative notions of *femininities* and *masculinities*, of *women* and *men* (Gendered Innovations, 2020, online).
- (2) Racial stereotypes: these reflect the beliefs that all members of the same race share given characteristics (Plous and Williams, 1995). These attributed characteristics are usually negative (Green, 1998).
- (3) Religious stereotypes (Bloomfield, 2016).
- (4) Color stereotypes: these represent the beliefs that colors contain a remarkable linkage with gender-related stereotypes. In particular, blue is for girls, as it is seen as more “dainty,” while pink used to be associated with boys, as it is seen as the stronger color. This is the “blue versus pink stereotype” that is driving gender inequality (Aaronson, 1970).

- (5) Age stereotypes: these are beliefs concerning features of the aged population. They could be refined and amplified across the life span of people and could be manifested in both positive (e.g. wise and generative) and negative forms (e.g. unproductive and forgetful) ([Chan et al., 2020](#) online).
- (6) Sex and sexual orientation stereotypes: these refer to the physical differences between people who are male, female or intersex and how people identify themselves, reflecting people's perceptions of the social roles of men and women in society.
- (7) Stereotypes about parents: these are beliefs about the role that typically should be attributed to parents, distinguishing between the role of the mother and the father.
- (8) Stereotypes related to substance addiction: these are social models that project on people who use drugs (by choice or to treat illnesses), which entail physical and psychological traits and behavior often judged as negative ([Movahedi, 1978](#)).
- (9) Stereotypes about teachers and evaluators.
- (10) Stereotypes related to genetic characteristics and different physical abilities.
- (11) Stereotypes related to the professions.

Stereotypes, as Lippman himself acknowledged, permeate social systems, regardless of the age, gender and profession of individuals; however, they are particularly active in organizations where judging and deciding on the basis of stereotypes often leads to *unethical* and usually even *incorrect* behavior. Lippman thus identifies the cultural origin of stereotypes within *organizations* as follows

The members of a hierarchy can have a corporate tradition. As apprentices they learn the trade from the masters, who in turn learned it when they were apprentices, and in any enduring society, the change of personnel within the governing hierarchies is slow enough to permit the transmission of certain great stereotypes and patterns of behavior. From father to son, from prelate to novice, from veteran to cadet, certain ways of seeing and doing are taught. These ways become familiar and are recognized as such by the mass of outsiders ([Lippman, 1922](#), p. 145).

Stereotypes are deeply embedded within social institutions and the wider culture ([Gendered Innovations, 2020](#), online); however, they are particularly harmful in the organizations in which they are widespread because every time an assessment or selection of individuals is necessary – for example, as in the case of personnel selection processes, career advancement, the assignment of tasks and the recognition of rewards and punishments – they can lead to *discriminatory* behavior. Stereotypes, however, hardly ever act as individual mental models, but together they can influence the actions of individuals. [Holvino \(2010\)](#) explicitly states that, in studying organizations, an intersection is observed in the action of stereotypes of race, gender and class.

I therefore want to present stereotypes as a form of *harmful mental model* that produces less efficiency and reduces performance, thereby causing a waste of human energy dedicated to work by reducing or slowing down the productivity of human labor ([Section 6](#)). In addition, stereotypes produce other relevant distortive effects because they lead to *discrimination* in society, communities and organizations, making nonrational distinctions between people based on the groups and classes they belong to ([Wikipedia, Discrimination, 2020](#)) and to the point of engendering individual and social violence ([Locke and Richman, 1999](#)).

Discrimination can produce *harmful social effects* that concretely appear *ex post* when the discriminating activities produce their results; however, discrimination becomes a real individual and social problem when it is perceived as an *injustice*, so much so that it can generate even permanent psychological effects ([Dion, 2002](#)). It must be kept in mind that when gender, race, ethnicity, age, sexual orientation, physical appearance and social class

discriminations exist (Carr and Friedman, 2005; Williams *et al.*, 1997), they are often not directly observable but manifest themselves only indirectly, in the form of ex post inequality across groups of the population (Klumpp and Su, 2013).

There are many types of discrimination. Many derive from, or are linked to, the action of stereotypes, while others refer to nonstereotyped variables, such as those associated with body weight or height, considered variables that create disadvantage in selections for particular jobs (Carr and Friedman, 2005); this discriminatory effect can be observed in almost all-important areas of activity, including education, sports and health (Pascoe and Smart Richman, 2009). I am aware that discrimination does not focus on the action of a single stereotype but often involves several stereotypes jointly, producing an effect that Collins (2015) and Collins and Bilge (2020) defines as “intersectionality.”

The term intersectionality references the critical insight that race, class, gender, sexuality, ethnicity, nation, ability, and age operate not as unitary, mutually exclusive entities, but rather as reciprocally constructing phenomena (Collins, 2015, p. 1).

Finally, it should be noted that discrimination is one of the factors that leads to prejudice, a preconceived opinion that is not based on reason or actual experience (Nelson, 2009; Dovidio and Jones, 2019) but on stereotypes (Cherry, 2020, online).

For this reason, Senge (1990), in studying the “learning organization” and developing organizational learning processes (Section 1), considers fundamental the mental models discipline and, while not explicitly mentioning them, stereotypes. He believes that this discipline must encourage, on the one hand, “reflection” on the stereotypes that influence our assessments and choices, and, on the other hand, the “awareness” of the injustices that stereotype-driven action can cause by making organizational action less efficient when valid individuals are excluded or penalized, without an objective selection process, solely because they fall within our stereotype models.

4. The maintenance and spread of stereotypes is due to a combinatory system

The combinatory system model shown in Figure 1 describes why stereotypes are so widespread and rooted in individuals, society and organizations that they are maintained over time. Some pass from one generation to another, often becoming stronger in consensus and intensity. To make the model more easily understandable, it is useful to recall the logic of combinatory systems.

In short, a combinatory system is a social system formed by a group of similar individuals or agents, not organized by hierarchical relations or interconnected by a network or tree relationships, that produce an analogous “microbehavior” that can generate some observable “microeffects” of accumulation, diffusion and order. Combined together, the “microbehaviors” produce a “macrobehavior,” which can produce an observable “macro-effect,” which, in turn, conditions the subsequent microbehavior of the individuals. Thus, on the one hand, the “macrobehavior” of the system, as a “collectivity”, derives from the combination (appropriately defined) of the analogous “microbehaviors” of the individuals (from which the name combinatory systems derives); on the other hand, the “macrobehavior” determines, conditions or guides the subsequent “microbehaviors.” This reciprocal relation can be defined as “micro–macro feedback”, and it can operate for a short time or for years or centuries (for a detailed description, see Mella, 2017). To understand how this operates, we must recognize that the “micro–macro feedback” requires the contemporaneous presence of “necessitating” and “recombining” factors. A necessitating factor can be defined as any element – a constraint, rule, condition, law, conviction, etc. – that “pushes” or “obliges” each individual in the collectivity to adapt his/her microbehavior to the macrobehavior of the system. The stronger the necessitating factor is, the more the microbehavior becomes activated. In combinatory systems made up of persons, the necessitating factors often originate from conscious motivations: necessity, convenience,

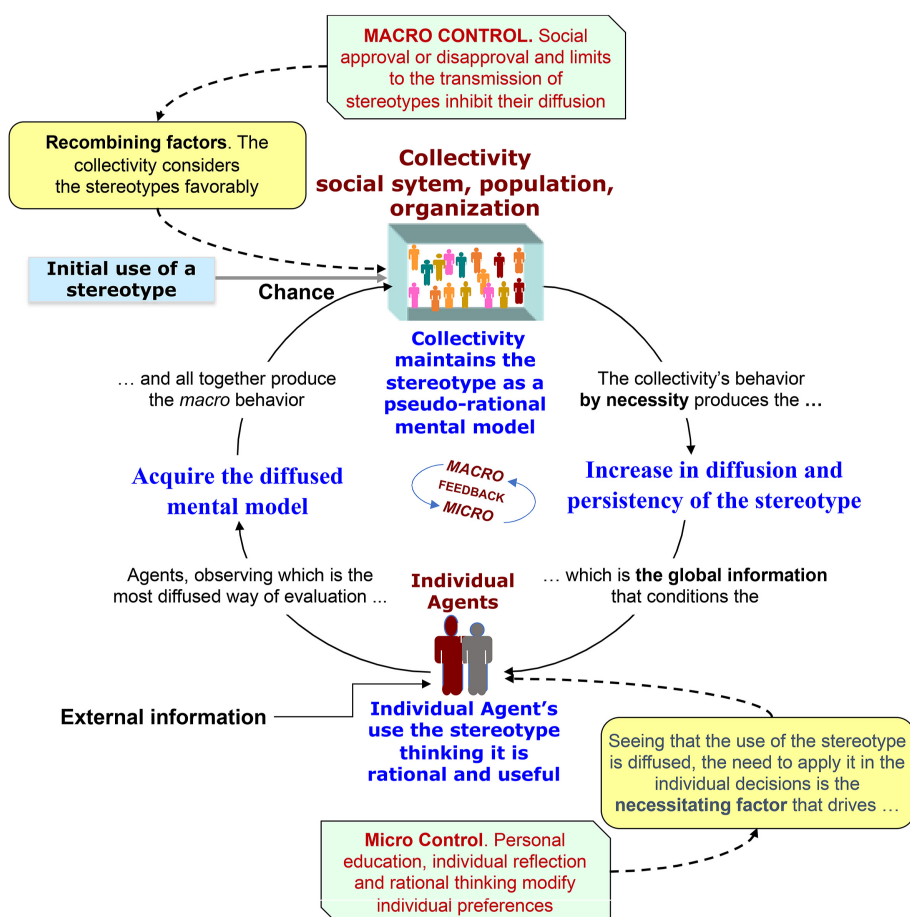


Figure 1.
Model of the
“maintaining the
stereotype” system

opportunity, the desire not to be inferior and so on. At other times, these factors can also be “natural” and act unconsciously since they derive from the cultural heritage of the individuals. The existence of one or more necessitating factors is indispensable though not sufficient; it is also necessary for the system – through some recombining factor (rule, convention, algorithm, nature of the environment the agents operate in, etc.) – to be able to recombine the microbehaviors (or microeffects) to produce the macrobehavior (or macroeffect) which, through the micro–macro feedback, can act on the necessitating factors. The model in Figure 1 illustrates how this logic can be applied to the spread and maintenance of gender stereotypes.

The gender stereotype, along with stereotypes of race and religion, is not only widespread at the workplace and in organizations but also in educational systems and regarding student performance. Such stereotypes (as well as many others equally as dangerous) can lead to significant inequalities, generating bullying, hatred and conflicts among young people with often fatal outcomes because it causes disidentification with school and sport, and the practices that may reduce these negative effects are difficult to implement (Steele, 1997, p. 613).

The combinatory system described in the model in Figure 1 allows us to understand how stereotypes are spread and maintained. Why does an individual adopt a stereotype?

Why are there dozens of stereotypes in the same social system, and why are they maintained? Stereotypes, gender stereotypes in particular, are so widespread, pervasive and resistant to control that it is natural to wonder what mechanism favours their existence and perpetuates them within a social group or population. First of all, we must remember that stereotypes are a component of social culture and are handed down, by use and example, from generation to generation. It is clear that the maintenance over time of stereotypes to evaluate, judge and decide the action of individuals is “regulated” by the typical action of a *combinatory system of diffusion*, according to which individuals *need to apply* stereotypes because they are *applied* in the social system in which they live; however, such individual behavior, *recombined* over time and space, stokes the use of such models in the social system. A typical *micro–macro feedback* is produced: on the one hand, the social system conditions individuals to adopt stereotypes, while on the other hand, it feeds on the use that individuals make of them, as shown in [Figure 1](#).

We are not interested in knowing when and why a certain stereotype arose among the members of a certain community, although one thing is certain: if “by chance” an individual, consciously or without reflection, is conditioned by a stereotype and *thinks he/she is in the right*, then “by necessity” he/she will transmit this way of thinking to his/her descendants, and this *microbehavior* will trigger the *macrobehavior* characterized by the perpetuation of the stereotype, which can be viewed as the *macroeffect of preserving the mental model*. The macroeffect, in turn, conditions the individual’s microbehavior illustrated in the model. This logic is summarized in [Figure 1](#), from which we can easily derive the rules for the heuristic model that describes this system

- (1) Microbehavior and necessitating rule
If you see that everyone judges according to a certain criterion, and you do not realize that they are following a stereotype, but parents and friends seem to believe in the model, then you will find it useful or necessary to conform to that common criterion of judgment and transmit this model to future generations, without being aware of perpetuating a stereotype.
- (2) Macrobehavior and recombining rule
If there are no interventions by the educational structures of the community to counter it, the stereotype will be widely adopted in evaluating and decision-making behavior, persisting and being transmitted unchanged at the collective social level.
- (3) Micro–macro feedback: chance and necessity
A stereotype is thus the result of the combination of past *microbehavior*, but it also conditions future behavior. A new stereotype, or a variant of a previous one, can be introduced in a given territory “by chance”, but when it spreads to a sufficient number of individuals, the community maintains the use of the stereotype “by necessity” through its transmission to its members and to newborns.
- (4) Strengthening, weakening and control actions
Social inertia toward reflection and change, together with the presence of situations in which the application of stereotypes appears to be successful as a quick criterion of evaluation and decision-making, considerably strengthen the system. The inclusion in schools of antidiscrimination education and the creation of rules of procedures and tools against discrimination due to stereotypes weakens the spread and strengthening of these mental models.

For reasons of space, it is not possible to present here a summary of combinatory systems theory. The reader is referred to Piero Mella, *The Combinatory Systems Theory* (Springer, 2017), for more detailed information on these kinds of systems.

5. Gender stereotypes and gender discrimination effects: the tournament theory model

Among the different types of stereotypes, I would like to focus on the gender stereotype, which “defines” how individuals of different genders “must” possess different characteristics that clearly differentiate them, thereby creating “expectations” about attitudes, roles, work, tasks, pay, etc., which should differ among individuals solely because a person is biologically a man or a woman (Hamilton and Trolie, 1986). Below are some examples of this stereotype

- (1) Women are emotional, while men are rational;
- (2) Men are strong and do all the work;
- (3) Women are not as smart as men;
- (4) Women cannot do as good a job as men can;
- (5) Girls are only concerned with physical appearance;
- (6) Women are good in the kitchen, while men at maintenance work;
- (7) Women love art and literature, while men love math and science;
- (8) Love is a female aspiration, while career is a male aspiration.

Gender stereotypes, and the discriminating opinions that characterize them, are not only present in the modern world but rooted in history, and over the centuries, they have produced a discriminating hierarchy between the status of a man and that of a woman (Schiebinger, 2014).

In brief, we argue that sex differences and similarities in behavior reflect gender role beliefs that in turn represent people's perceptions of men's and women's social roles in the society in which they live. In postindustrial societies, for example, men are more likely than women to be employed, especially in authority positions, and women are more likely than men to fill caretaking roles at home as well as in employment settings. Men and women are differently distributed into social roles because of humans 'evolved physical sex differences' in which men are larger, faster, and have greater upper-body strength, and women gestate and nurse children (Eagly et al., 2000).

According to the gender stereotype, the characteristics attributed to the male are the ability to act, courage, strength, self-affirmation, competence, manhood, individuality and a greater interest in social and public life, which is why only men were found in military careers until a few decades ago. The characteristics attributed to women, on the other hand, are emotional capacity, sociality, affectivity, emotiveness, communicative skills, altruism, interdependence, sweetness, delicacy, softness, the tendency to manage one's own intimate world, as well as the tendency to favor motherhood, childcare and family care (Diekmann and Eagly, 2000). Moreover, the gender stereotype attributes practicality, precision and independence to men, characterizing women as lacking courage and initiative and as being disordered; men are creative and innovative, while women are deemed in need of direction and continuous assistance at work.

Those who act according to the gender stereotype believe that the characteristics I have just mentioned are “truly” possessed by men and women and are led to *discriminate* between genders; this is manifestly irrational behavior because it is based on an *a priori*, unverified judgment that leads people to assert that, because of gender characteristics, *men and women cannot be on an equal footing in terms of rights, duties, roles, work, tasks and pay and must therefore be treated differently*. In countries where gender equality is well-established today, the stereotype's claims are clearly refuted by incontrovertible evidence: more and more women are becoming heads of state, government and institutions, making scientific and

technical discoveries and innovations, joining the military and achieving the same success as pilots and astronauts as men have achieved.

However, even today, the gender stereotype acts almost undisturbed in *organizations*, leading, for the most part, to favoring males at work by assigning them higher roles, competencies and salaries compared to those reserved for women. This normally creates a *distortion in corporate efficiency* in that the "talents" of women who are not properly valued are lost. Employees who have been discriminated against can show lower their self-esteem, and being less motivated, they will reduce their commitment to work and thus perform more poorly, thereby reinforcing the supposed kernel of truth contained in the stereotype, as indicated in the model in [Figure 2](#).

According to *tournament theory*, proposed by [Lazear and Rosen \(1981, 1990\)](#) hierarchical levels and salary differences should be based on productivity or merit; however, when evaluators are influenced by *stereotypes*, especially of gender or race, the hierarchical and economic differences are not based on the actual relative differences among individuals who are specifically evaluated so that the influence of stereotypes distorts the rationality of choices and leads to a waste of resources ([Gilovich et al., 2002](#)), thus also favoring gender discrimination in the organization. If the tournaments are biased, due to the influence of gender stereotypes, the odds of winning differ by gender, other things equal. "*The higher the influence of stereotypes, the lower the female probability of winning, and the wider the unexplained gender pay gap*" ([Castagnetti and Rosti, 2013](#), p. 633).

In addition to using [Figure 1](#), the latter effect can be better represented by the *success to the successful* archetype ([Mella, 2012](#), Sect. 4.9), as indicated in [Figure 3](#).

The **R1** loop highlights the effect of male workers' privileges: when role and pay choices are driven by gender discrimination, workers are usually given a high-level role resulting in

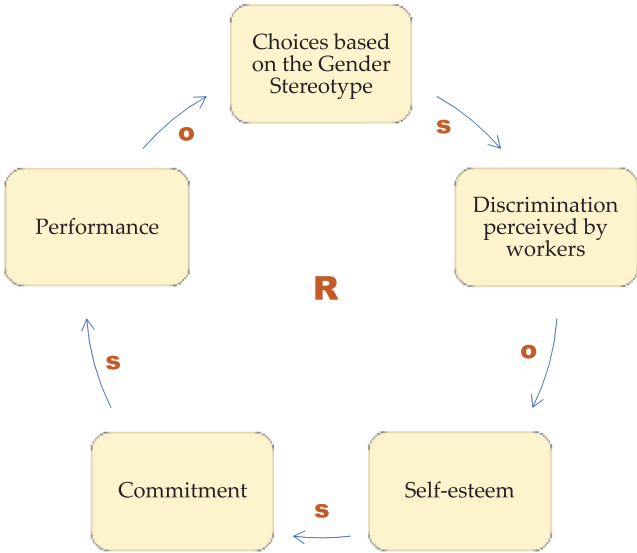
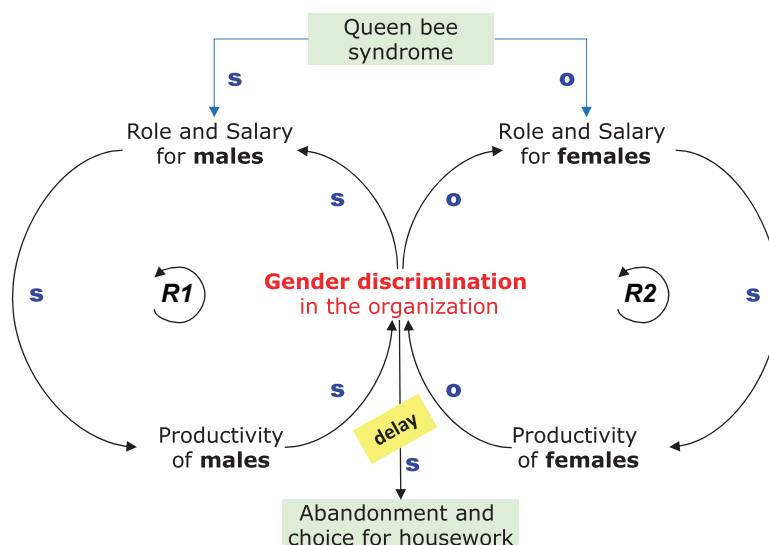


Figure 2.
Gender discrimination
and distortion of
business efficiency

Note(s): "s" means "same direction/sign variation; "o", opposite-direction/sign variation; "R" indicates a "reinforcement" loop because it tends, over time, to amplify the variables indicated in the boxes. See Sect. 2



Note(s): Figure 2 and Sect. 2

Figure 3.
Gender discrimination
in the organization
(according to
Tournament Theory)

high pay; this improves worker satisfaction, which leads to higher productivity and evaluators to believe discrimination is correct. The **R2** loop reinforces this belief as lower roles and low worker pay lead to dissatisfaction and a lack of effort to improve productivity. The combined effect of the two loops justifies, maintains and reinforces in the evaluators the *irrational idea* that gender discrimination is the result of an actual difference in productivity levels linked to gender and not to the discriminatory action of the stereotype described by the model. The use of stereotypes by the evaluator results in an increase in discrimination, as he/she judges people by the group to which they belong.

The simplicity of the model in [Figure 2](#) helps to highlight two other effects of discrimination in organizations: on the one hand, women workers who can no longer tolerate discrimination prefer to give up work and devote themselves to working for the family, and on the other hand, the *queen bee syndrome* occurs: a kind of reverse discrimination according to which women who have succeeded in career advancement and have achieved a position of authority treat women who work under them more severely and critically than they do male subordinates, thereby hindering the potential professional growth of other women and favoring male employees to ensure there is no other women in positions of power.

6. The role congruency theory, the glass ceiling theory, self-fulfilling prophecies and the Pygmalion, Golem and Galatea effects

Another important effect of gender discrimination is the *slowness* of the career paths of women in organizations of all kinds, compared to the *speed* at which *men* advance in their careers. In fact, the gender stereotype can also be “masked” by the idea that there must be a de facto congruence between certain types of work and roles and the gender of workers. [Eagly and Karau \(2002\)](#) examined the role congruency theory, according to which women and men are expected to behave in ways that “match” their gender roles: that is, the idea of role congruity. According to this theory, certain roles can be seen as stereotypically masculine or feminine ([Konrad et al., 2000](#)), leading perceivers to expect a particular role-congruent set of attitudes and behavior. Therefore, the behavior of an individual performing a job that is not

congruent to his/her role is negatively assessed, which demotivates the individual and reduces his/her performance. According to the *glass ceiling theory* (Powell and Butterfield, 1994; Powell *et al.*, 2002), there is an invisible barrier that keeps women from rising in a hierarchy (Ferber and Nelson, 1993); therefore, *as the prestige of a job increases, the share of women in that position decreases*.

According to role congruency theory, there are three stereotypes, or factors, that lead to discrimination in the career advancement of women (Heilman and Parks-Stamm, 2007)

- (1) “THINK MANAGER, THINK MALE” (for the role of secretary the opposite is true) is a typical gender stereotype that leads to discrimination in the evaluation and selection of managers, setting roadblocks in the way of women with regard to leadership positions (Schein *et al.*, 1996) and revealing a negative feature of the cognitive process that distorts the use of information (Hudak, 1993), thereby producing systematic errors in the results obtained in both production organizations and at school (Biernat and Kobrynowicz, 1997; Comer and Drollinger, 1997).

A role congruity theory of prejudice toward female leaders proposes that perceived incongruity between the female gender role and leadership roles leads to 2 forms of prejudice: (1) perceiving women less favorably than men as potential occupants of leadership roles and (2) evaluating behavior that fulfills the prescriptions of a leader role less favorably when it is enacted by a woman. One consequence is that attitudes are less positive toward female than male leaders and potential leaders. Other consequences are that it is more difficult for women to become leaders and to achieve success in leadership roles. Evidence from varied research paradigms substantiates that these consequences occur, especially in situations that heighten perceptions of incongruity between the female gender role and leadership roles (Eagly and Karau, 2002, p. 537).

- (2) “MATERNAL WALL”, which consists in judging *mothers* as being less dedicated to work, less competent and less suitable for recruitment or promotion than nonmothers, thereby proposing hierarchical levels and lower wages for mothers (Crosby *et al.*, 2004).

While some women stand nose pressed against the glass ceiling, many working mothers never get near it. What stops them is the “maternal wall.” Where mothers are concerned, coworkers and bosses often perceive a trade-off between competence and warmth. [...] Women who have been very successful may suddenly find their proficiency questioned once they become pregnant, take maternity leave, or adopt flexible work schedules. Their performance evaluations may plummet and their political support evaporate. The “family gap” yawns: An increasing percentage of the wage gap between men and women is attributable to motherhood (Williams, 2004, p. 26).

- (3) “FAMILY RESPONSIBILITIES WALL”, which concerns the disadvantage women face in career advancement. Women’s careers also depend on the distribution of time and effort devoted to home care and to their careers (Greenhaus and Parasuraman, 1999). Therefore, only if women have a comparative advantage in caring for and raising children, and/or if they suffer discrimination in business activities, will the traditional division of gender work arise. In this case, domestic responsibilities will lead married women to reduce the effort provided and the time spent at the workplace, thus avoiding the commitment necessary for the top roles. In addition, given that all the activities carried out, both at home and at work, are either *effort-intensive* or *time-intensive*, women who devote a lot of time to *effort-intensive* work at home will tend to economize on their energy use by looking for jobs that are not effort-intensive, while women who devote little time and effort to domestic work will behave in the opposite manner.

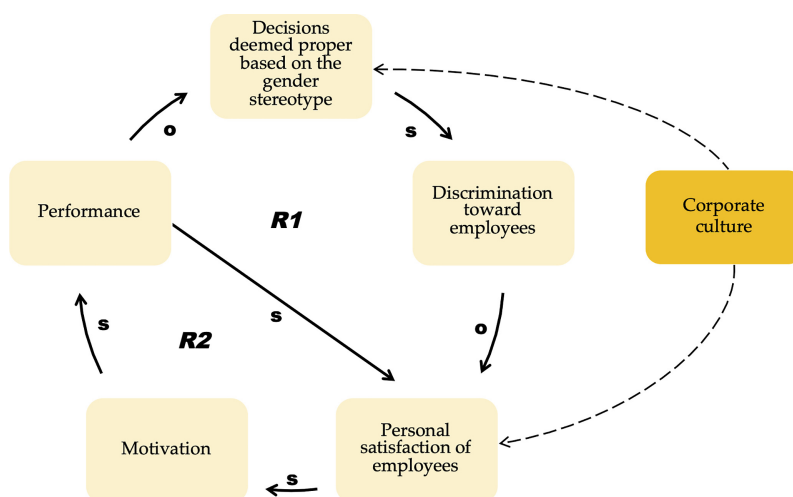
People who apply stereotypes unconsciously tend to give preference and store information that confirms their stereotypes over information that contradicts them, deeming such information to be true only because it corresponds to their expectations. Stereotypes can influence behavior; thus, the error is not corrected *because the prophecy is self-fulfilling*. A model for interpersonal *self-fulfilling prophecy* is represented in Figure 4, which supplements the Pygmalion effect.

An example of a self-fulfilling prophecy is when the "selection process" is repeated over time based on the gender stereotype and produces the *Pygmalion Effect*: stereotypes condition the evaluation and selection processes in such a way that the individual who most reflects the preconceived judgment (stereotype) of the evaluator is chosen (Rosenthal and Jacobson, 1968; Eden, 1984). In addition, if the evaluator is prejudiced regarding the abilities of the person being evaluated and expects the latter's performance to match these expectations.

The *Pygmalion effect* (loop R1 in Figure 5) can be contrasted with the so-called *Galatea effect*, which produces a *self-fulfilling prophecy* for the individual who evaluates himself/herself (loop R2 in Figure 5). This effect occurs when an individual's high expectations of himself/herself translate into high performance, and the results achieved influence self-efficiency: the more such individuals are convinced of their ability to achieve a goal, the more likely it is to happen.

The model also highlights the role of *corporate culture* in influencing supervisor/evaluator expectations and employee expectations. In fact, the corporate culture is formed by unwritten rules, values, procedures, routines and styles that influence people's expectations, thought, and behavior in an organized environment; it is the result, among other things, of the *core values* and *business principles* adopted by managers, of the *approach to human resource management*, of the *corporate climate* and of the *operating style*.

The *Pygmalion effect* can also be associated with the relationship between the *expectations* the superior, or supervisor, has regarding the performance of the subordinates and the latter's *level of performance*. The higher the expectations, the more likely the subordinate will show a high level of performance. However, the opposite can occur: the lower the supervisor's



Note(s): see Figure 2 and Sect. 2

Figure 4. Gender discrimination and diminished self-esteem of an employee discriminated against based on the role congruency theory

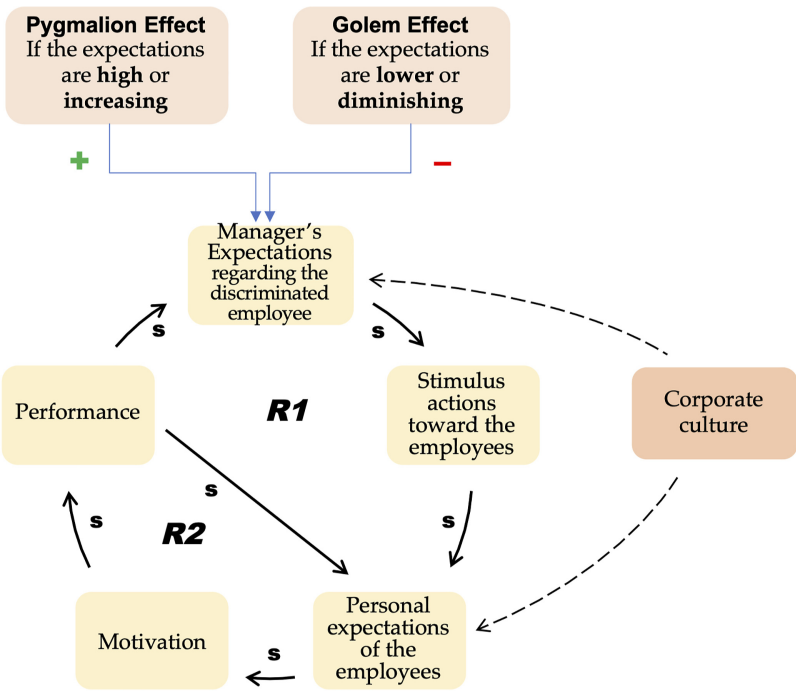


Figure 5.
Gender discrimination and the influence of expectations on the performance of employees who suffer discrimination

Note(s): see Figure 2 and Sect. 2

expectations, the more likely it is that the subordinate, feeling discriminated against and underappreciated, will produce a lower performance level. This variant, which “reverses” the Pygmalion effect, is also known as the *Golem effect* (Babad *et al.*, 1982; Reynolds, 2007). The two effects can be represented in the model in Figure 5.

Figure 5 can be interpreted in two directions. If the manager/supervisor’s expectations of the employee are high or increasing, then the model produces the *Pygmalion effect*: an improvement in the performance of employees as they increase the expectations of the manager results in an increase in the positive stimuli transmitted to employees, which leads to an increase in the latter’s personal expectations because they feel appreciated and an increase in their motivation and improved performance. This result confirms to the manager that he has made the right choice (self-expected prophecy), thereby increasing expectations even more. If, on the other hand, the expectations of the manager/supervisor are low, or are reduced, then the model produces the *Golem effect*: the incentives for the employee are reduced, which leads to lower motivation and performance so that the manager/supervisor once again believes that discrimination has proven to be the right choice.

7. Negative effects that evaluations based on stereotypes have on productivity

The effects described by the models in the two previous sections not only affect individual workers but can produce significantly more general effects on company productivity and, from a broader perspective, also on the productivity of a production system. For this reason, we must consider human labor not as the simple execution of acts and processes but as “energy” strenuously delivered to obtain productive results, goods and services.

As Carlo Cipolla demonstrated in his masterful and detailed essay, *The Economic History of World Population* (1962), from an economic point of view, the history of humanity can be interpreted as a man's continuous search to optimize the application of "energy" (that is, of "labor") to production (and also to consumption) to achieve the maximum production with the same amount of energy/labor to obtain goods and services to meet mankind's needs and aspirations and/or minimize the amount of (strenuous) energy needed to obtain a given amount of production.

Carlo Cipolla offers a synthesis of the history of mankind as an evolution of its relationship with the "energy" dispensed to survive from prehistoric times, when man was nothing more than a predator who acquired knowledge and skills for hunting and fishing, to when, during the eighth millennium B.C., he became a farmer and breeder and learned to select the stock of plants and animals for his own use and to optimize the "yield" with a lower quality of energy/labor. Even children could graze a flock or reap the fruits of the Earth. Man also learned how to enhance his energy/labor by combining it with inanimate energy sources (wind and water currents) or domestic animals. With the Industrial Revolution in the 18th century and the invention of the steam engine (and its evolution), man took the further step of supporting and enhancing his work with the aid of steam power and mechanical energy. To this end, he designed and built machinery, plants and equipment of all kinds that "enormously" increased the products obtained with the same amount of energy/human labor employed and searched for the most effective energy sources. In the modern era, computer science and artificial intelligence has made the use of nonhuman resources for production more widespread.

To measure the efficiency of the energy/labor used to obtain given quantities (and a given level of quality) of production, the term "productivity" is now used, which (very briefly) quantifies the relationship between *the output* of production processes, products and services, and the *labor inputs* necessary to obtain them. The European Commission communication to the European Council and Parliament of 21 May 2002, the subject of which was *Productivity: The Key to Competitiveness of European Economies and Enterprises*, provides this definition

In formal terms, labour productivity is the quantity of labour required to produce a unit of a specific product. In the macroeconomic context, labour productivity is measured as a country's gross domestic product (GDP) per capita of employed population. Productivity growth depends on the quality of physical capital, improvements in the skills of the labour force, technological advances and new ways of organising. Productivity growth is the principal source of economic growth (COM, 2002).

If we let " πL " stand for the "average productivity of labor (L)" (in a period), then the COM definition, indicating the *efficiency in the use of the labor factor*, can be quantified by the ratio [1]:

$$\pi L = \frac{QP}{QL} = \frac{Qp}{qL \ QP} = \frac{1}{qL}$$

where QP indicates the volume of production obtained by a worker, a company or an economic system, QL is the quantity of human labor used to obtain QP and qL is the unit quantity of human labor needed for a unit of product (for simplicity, the time period has not been specified).

Since productivity shows a tendency to increase over time (Mella, 2018), in order to understand what this productivity dynamic depends on, even the first economists tried to identify some *determining factors* or *drivers of productivity*, which, despite their variety, can be grouped into few categories. Although these categories interact, they must be kept separate to facilitate the explanation derived from Adam Smith's famous example of the manufacture of pins to illustrate the advantages of the division of labor.

This great increase in the quantity of work, which, in consequence of the division of labour, the same number of people are capable of performing, is owing to three different circumstances; first, to the increase of dexterity in every particular workman; secondly, to the saving of the time which is commonly lost in passing from one species of work to another; and lastly, to the invention of a great number of machines which facilitate and abridge labour, and enable one man to do the work of many (Smith, 1776, p. 5).

We can expand Adam Smith's analysis by observing from ratio [1] that increasing productivity generally means (Mella, 2018)

- (1) increasing the quantity (and/or quality) of goods produced to meet the needs and aspirations with the same amount of labor employed in production (increase QP in the numerator while QL in the denominator remains constant);
- (2) reducing the amount of labor required to produce a given quantity (and quality) of products obtained (QL in the denominator is reduced while QP in the numerator remains constant) and
- (3) a mix of the previous effects.

Despite their variety, the *productivity drivers* capable of producing the effects mentioned above can be grouped into the following classes

- (1) *Passive drivers*: these affect the quantity of production (numerator) given the same amount of labor employed; there is only one passive factor of productivity: "natural or artificial" *fertility*. This is the driver that characterized the increase in productivity during the agricultural and breeding revolution 10 millennia ago;
- (2) *Active drivers*: these affect the amount of energy/labor needed for production; these drivers reduce this quantity (or improve the quality of performance) given the same level of fertility. There are three active drivers of productivity:
 - *Skill*: a skilled worker is able to produce the same quantities of goods in less time than nonskilled workers can; conversely, by applying the same amount of labor as other workers do, they are able to obtain a greater quantity of products;
 - *Equipment*: equipment extends the capabilities of the human "hardware" represented by the body and its limbs and brain, reduces fatigue and danger, and rewards skill. With the same amount of labor, the amount of production obtainable by using equipment increases enormously.
 - *Specialization*: the *functional division of labor* is characteristic of the organization, which means that each worker in the organization supplies *specialized* and *equipped* labor for a *specific activity*. By specializing in their work, individuals in an organization further increase productivity; *specialization* requires choosing workers with greater skill and potential, fostering coordination and cooperation among the operators themselves and forming organizational hierarchies with career opportunities;
- (3) *Endogenous* or *psychological* drivers: these are the psychological conditions that "push" people to provide labor to a given organization. We can distinguish between
 - *Motivation*: man is willing to provide his labor only if he has adequate motivation and expects his needs or aspirations to be satisfied. In the past, the main motivation was represented by remuneration, in the form of both wages or salary and profit; today, the motivation connected to monetary remuneration is joined by

others of an intellectual nature: performing a job that is satisfying is often preferable to a better paid but boring job;

- *Fulfillment*: motivation encourages people to begin to work, but the *initial motivation* must be followed by *fulfillment*, that is, the satisfactory achievement of motivations.

This lengthy introduction leads us to a rapid conclusion. The negative influence of choices made under the influence of stereotypes is immediately evident, especially the effect of “gender stereotypes” on productivity. There are two drivers that are immediately influenced: the *skill* driver and the *psychological* drivers of *motivation* and *fulfillment*.

If entry into an organization were guided only by *skill* and by the potential of the worker, there would be an increase in efficiency and an increase in collective productivity. The task of evaluating candidates is now implemented through different *levers*: technical assessments, with objective tests evaluated anonymously by commissioners outside the organization, the evaluation of a candidate’s curriculum vitae, recruitment preceded by a trial period, specific training courses provided to candidates, etc. Unfortunately, even today these *levers*, although widely applied, are not sufficient to effectively counteract the action of *gender stereotypes*; in schools, universities, hospitals, public offices, military bodies and even regarding professional activities, a candidate’s assessment can still be affected by gender stereotypes. The consequence is that many candidates with skills that are useful for the organization risk losing out to less-skilled individuals solely based on gender differences. Today, the number of women in a job once considered “male” – the armed forces, firefighters, the police, etc. – is constantly increasing. At the same time, professions once considered predominantly “female” – kindergarten and elementary school teachers, the care of the elderly, beauticians, etc. – have seen an increase in male workers. Keeping professions separate because of gender discrimination clearly results in a “waste” of skills, knowledge and potential and therefore in a waste of productive “energy”.

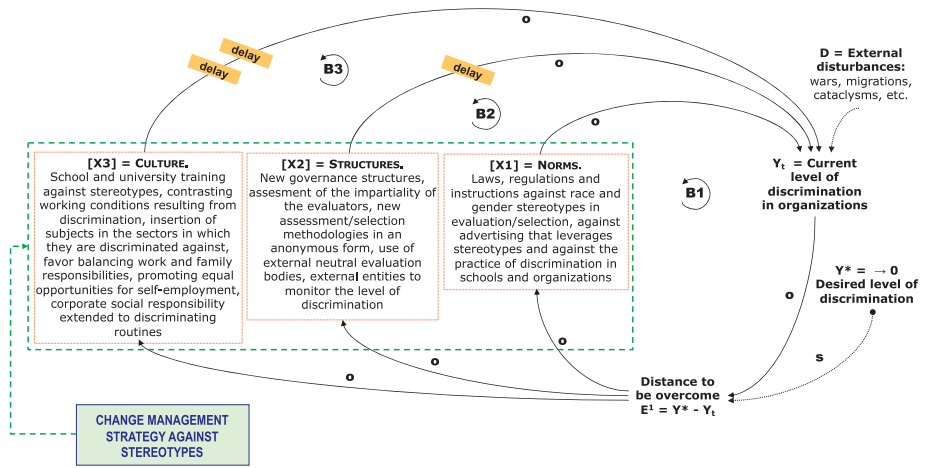
As the models in the previous sections reveal, gender discrimination acts above all on the *psychological drivers*, *demotivating workers and reducing their efficiency, which leads to a waste of productive “energy.”* There is no need here to continue with a more detailed explanation.

8. The logic of PSC theory for the control of stereotypes

The combinatory systems model examined in [Section 4](#) describes why the gender stereotype, along with stereotypes of race and religion, are so widespread and rooted in individuals, society and organizations and why these dangerous mental models are maintained over time, passed down from one generation to another with an increase in consensus and intensity of belief. The stronger the stereotype, the more incisive its effects will be, and the longer they are likely to last. The immediate recognition of certain stereotypes means they are widely used in the production of effective advertising and in sitcoms ([Bauer et al., 2018](#); [Akestam et al., 2021](#)).

Since discrimination based on stereotypes of gender and, in particular, of religion and race are typically the result of widespread “cultural distortion,” the fight against stereotypes is above all a battle for *change in cultures* that judge, evaluate and select on the basis of *unfounded prejudices*. Controlling the effects of discrimination due to the social spread of stereotyped judgments imposes a social change that leads to the modification of the *collective culture* and the abandonment of stereotyped thinking. A review of the vast literature reveals some themes that remain unconnected and underdeveloped in the research on change. We draw upon the systemic approach to further theorize about the nature of change and how to deal with it, and we focus on the underresearched role of culture in the context of organizational change, addressing how organizational culture does, in fact, contribute to organizational change, and how a change process is implemented.

Figure 6.
Change management
strategy against
discrimination due to
stereotypes, and to
gender stereotypes, in
particular. The
PSC Model



Note(s): see Sect. 2

To understand the conditions for such a change, we can refer to *PSC theory* (Mella and Colombo, 2012; Mella, 2014), according to which all *collective change* takes place over three stages that require different modus operandi and implementation times (Figure 6)

- (1) **[P]** *rapid change*, produced in theory by “rules” requiring changes in “operational and decision-making processes” (e.g. the imposition of quotas for women in all collegiate bodies, the modification of job competitions by imposing gender equality, etc.);
- (2) **[S]** *structural change establishing appropriate structures in the social or organizational context* according to rules, regulations and instructions that oppose discrimination in schools and organizations and in advertising messages of all kinds (e.g. creating specific periodic audits to detect discrimination in organizations; establishing appropriate chambers in the courts to discuss only those cases related to discriminatory behavior; opening “listening centers” throughout the country; imposing competitions for job promotions with externally appointed commissioners; making participants in job selections as anonymous as possible, etc.). Since organizational adaptation takes time, structural change, though necessary, acts more slowly;
- (3) **[C]** *cultural change, which makes regulatory and structural changes permanent* but takes a long time and, in the case of gender and race stereotypes, may require more than one generation (Plous and Williams, 1995). If the evaluation distortions result from cultural factors, it is necessary to act against these stereotypes and produce a cultural change to identify and optimally allocate “talent.” The identification of talent has social value since, if not allocated properly, actual output will fall short of potential output. As we have seen, according to tournament theory (Lazear and Rosen, 1981), social welfare is greater if the most gifted occupy the top positions.

The model in Figure 6 illustrates a change management strategy to fight stereotypes. It contains some abstract indications that need to be adapted to real-world situations.

Regulatory intervention (rapid change, “P”) and the *creation of suitable structures* (structural change, “S”) are necessary to initiate (or even force) “concrete actions” to combat discrimination, i.e. *specific projects* aimed at removing obstacles that, in fact, prevent the full achievement of “equal opportunities”. Particularly important in this case are concrete actions

based on the principle of ensuring “equal opportunities” to combat discrimination in organizations, since they can become a decisive way to change an individual’s perception of “career advancement” possibilities and avoid a propensity for employees to resign in the absence of equal opportunities. Being “aware” of the existence of widespread and well-entrenched discriminating choices represents the basis for initiating over time a real cultural change (“C”) toward “equal opportunities” in work and in relational life, counteracting discrimination with regard to gender, race and religion.

- (1) Among the “concrete actions” which require cooperation between the employer and the employees, the following are particularly necessary and useful:
- (2) Eliminate the de facto disparities between men and women (or disparities that result from other stereotypical characteristics), which manifest themselves in educational preparation and vocational training, access to work, career advancement, working life and periods of mobility.
- (3) Prohibit rules, procedures and routines that consider individuals of a given gender as unfit to play a given role or perform a given job (in the army, navy, in activities related to public safety, in sports, in scientific research, etc.).
- (4) Take concrete actions in line with the slogan “When this is a man”, which emphasizes the positive role of men working in areas traditionally considered feminine (and vice versa) in order to change gender roles and overcome stereotypes (teachers, nurses, beauticians, etc.).
- (5) Guarantee everyone the fair and inclusive use, without prejudice, of quality educational and university services, favoring postmaster specialization and professional practice.
- (6) Promote, through rational work organization and a system of public services, a balance between professional and family responsibilities that is fair to both sexes.
- (7) Guarantee and facilitate the right to maternity and childcare, with social assistance treatment for the family.
- (8) Encourage diversification in women’s professional choices and their access to self-employment, through specific conditions regarding access to financing.
- (9) Facilitate access to medical care, sexual health and financial resources.
- (10) Overcome discrimination arising from stereotypes in the organization of procedures and the distribution of workloads, with particular attention to prejudices relating to training, professional and career advancement, and economic treatment and salary structures.
- (11) Promote access to professional activities and sectors where women are underrepresented, in particular, in corporate and administrative bodies.
- (12) Of course, “concrete actions” must be extended to all individuals and not become a kind of “reverse discrimination”, i.e. preferential treatment only for the least represented gender (or with regard to another aspect of discrimination).

In addition to the “concrete actions” presented above, I would like to mention the “Agendas” provided by various countries and international organizations to eliminate gender disparities; in particular, the United Nations *2030 Agenda for Sustainable Development* (2020) “Goal 5: Achieve gender equality and empower all women and girls.”

Women and girls, everywhere, must have equal rights and opportunity, and be able to live free of violence and discrimination. Women's equality and empowerment is one of the 17 Sustainable Development Goals, but also integral to all dimensions of inclusive and sustainable development. In short, all the SDGs depend on the achievement of Goal 5 (SDG 5, online).

Nine targets are indicated, including

Ensure women's full and effective participation and equal opportunities for leadership at all levels of decisionmaking in political, economic and public life.

Ensure universal access to sexual and reproductive health and reproductive rights as agreed in accordance with the Programme of Action of the International Conference on Population and Development and the Beijing Platform for Action and the outcome documents of their review conferences.

Undertake reforms to give women equal rights to economic resources, as well as access to ownership and control over land and other forms of property, financial services, inheritance and natural resources, in accordance with national laws (SDG 5, online).

About education, also of note is the declaration relating to Goal 4 which is part of the [UNESCO's Education Agenda 2030 \(2016\)](#): Incheon Declaration and Framework for Action for the Implementation of Sustainable Development Goal 4: *ensure inclusive and equitable quality education and promote lifelong learning opportunities for all* (UNESCO, 2016, online).

9. Conclusion and further developments

This study has considered the action of stereotypes, *gender* stereotypes in particular, as especially harmful "mental models," highlighting the distortions they cause in society, groups and organizations since they normally produce a waste of productive energy that, if not compensated for by binding process automatisms capable of canceling the effect of the lower skills and motivation of workers and by a more intense use of machinery, can reduce the productivity of labor. I have followed a predominantly organizational and corporate approach, although the copious literature on stereotypes belongs largely to the area of *social psychology*. Using the "logic" and "language" of systems thinking (Mella, 2012), I have tried to present the theories and models that describe and interpret the distorting effects of organizational choices based on stereotypes rather than rational analysis. I have also tried to develop a combinatory systems model whose *modus operandi* can interpret the genesis and maintenance over time of stereotypes. Finally, using the PSC model, I have shown how stereotypes can be dealt with to attenuate or eliminate their effect through interventions on the social culture that incorporates the stereotypes themselves: by acting on the three wheels of change, highlighted in the PSC model, through legal provisions, control tools, and actions on the culture that exists in educational and social aggregative institutions, it is possible to change the prevailing culture so that it does not understand, or rather, does not accept, the influence of gender stereotypes and other discriminatory mental models.

There are three avenues for future research into the harmful role of stereotypes capable of causing discrimination and inefficient choices.

First, it is useful to remember that many studies on stereotypes demonstrate that those dangerous mental models act together, their dangerous effects "intersecting", to influence evaluations and choices: e.g. gender stereotypes, racial stereotypes and stereotypes related to genetic characteristics and different physical abilities or age, sex and sexual orientation stereotypes and stereotypes about parents (Holvino, 2010). It would be useful to investigate which sets of stereotypes form an effective "intersection" in different types of organizations, productive and otherwise. I have not delved into the problematic aspects of racial stereotypes – widespread on every continent – and the resulting negative effects related to racially biased decision-making, racial disparities and racial injustice. These aspects were well highlighted in

the Report of the [Working Group of Experts on People of African Descent \(General Assembly, 74th Session, 2019\)](#), Item 70 (b), and by [Kaschula et al. \(2012\)](#), as evidenced in the sites that Reviewer has pointed out to me, for which I thank him. I have, in fact, focused the object of my research on the action of gender stereotypes in organizations.

Second, it would be extremely interesting to study the functions, structures and actions of diversity management that productive and nonproductive organizations set up to manage diversity; specifically, which coexistence controls and interventions diversity management implements to *foster inclusive work* ([Jackson and Ruderman, 1995](#)), *improve organizational efficiency* ([Klarsfeld, 2010](#); [Scott et al., 2011](#)) and *combat the “stereotype threat”*, ensuring that persons who are discriminated against do not end up identifying with the stereotypes that have contributed to their situation and become convinced that the discrimination is fair and that they deserve the position they are in ([Roberson and Kulik, 2007](#)).

Third, it would be extremely useful to investigate economic and political measures that can facilitate the application of the PSC model and speed up the cultural change involving the abandonment of harmful mental models. The effectiveness of “concrete actions” should be more closely examined to identify those that can achieve the objectives of combating stereotypes more widely and in a shorter time.

Finally, it is essential to identify educational measures – in schools, associations, families, places of worship, etc. – that would be effective in *transforming stereotypes from harmful social models into models of reflection* useful in sparking curiosity towards the “culture of the other” to enhance diversity, recognize differences and gain awareness of everyone’s cultural identity, attributing to individuals the value they deserve and not one altered by a judgment based on stereotypes. With regard to *gender stereotypes*, for example, it should be emphasized that in organizations and social systems, *gender diversity* should be considered an opportunity and not as a discriminating factor and thus encouraged by avoiding harmful discrimination; this diversity, in fact, precisely because of the presence of the distinctive characteristics of individuals regardless of sex, can benefit the organization and lead to an increase in organizational and social performance.

Referring to the last point, among the many counterintuitive factors adduced to show that greater gender diversity is correlated to greater business performance, the following are particularly significant

- (1) The signal of an *improved organization* capable of producing *greater business and social value*. Some studies ([Adams and Ferreira, 2009](#)) show that the appointment of more women to executive positions is a positive signal that is sent out by the organization, since it indicates a greater focus on corporate governance; “*Women on boards: Not just the right thing [. . .] but the ‘bright thing’*” ([Brown et al., 2002](#)); and, at the same time, it is a sign that organizational performance is positive and that companies show higher profitability ([Carter et al., 2003](#)). Studies by McKinsey have demonstrated that higher returns on equity are achieved in companies with female directors compared to those without female directors ([McKinsey and Company, 2007](#)).
- (2) *Increased effort from a heterogeneous group and access to a wider source of talent*. Since women have higher educational outcomes and retention rates, their access to managerial careers increases the amount of talent in the organization ([Hillman et al., 2007](#)). It has been shown ([Thomas-Hunt and Phillips, 2004](#)) that greater heterogeneity in working groups improves average performance and social sensitivity. However, greater gender diversity must not lead to increased tensions and conflicts ([Homan et al., 2007](#)).
- (3) *A better mix of managerial skills*. In “Women Matter”, [McKinsey and Company \(2007\)](#) points out that there are differences in managerial style between men and women ([Singh and Vinnicombe, 2004](#)). There are nine criteria that define a good leader, and women

possess five of these behavioral criteria more often than men do. For example, women are better at defining responsibilities and training people, while men are more adept at making individual decisions and taking corrective actions in difficult situations.

- (4) *Improvement in corporate governance and greater attention to consumer preferences.* One study (Brown *et al.*, 2002) finds that boards of directors with three or more women perform better in terms of governance than do companies with only male board members, since greater gender diversity leads to a focus on clear communication with employees, prioritizing customer satisfaction and explicitly highlighting the uniqueness of the company (Krishnan and Park, 2005; Kramer *et al.*, 2006).

Recognizing stereotypes and their negative effects on organizational performance does not mean being able to control their harmful diffusion in all productive contexts; however, it makes us aware of the need to cancel their influence through "concrete actions" that achieve the cultural change suggested by the PSC model. By acting on the three wheels of change highlighted in the PSC model: legal provisions, control tools and actions on the culture existing in educational and social institutions, it is possible to change the prevailing culture so that society can understand, or rather accept, the influence of gender stereotypes and other discriminatory mental models.

References

- Aaronson, B.S. (1970), "Some affective stereotypes of color", *International Journal of Symbolology*.
- Adams, R. and Ferreira, D. (2009), "Women in the boardroom and their impact on governance and performance", *Journal of Financial Economics*, Vol. 94 No. 2, pp. 291-309.
- Åkestam, N., Rosengren, S., Dahlén, M., Liljedal, K.T. and Berg, H. (2021), "Gender stereotypes in advertising have negative cross-gender effects", *European Journal of Marketing*, Vol. 55 No 13, pp. 63-93.
- Babad, E., Inbar, J. and Rosenthal, R. (1982), "Pygmalion, Galatea, and the Golem: investigations of biased and unbiased teachers", *Journal of Educational Psychology*, Vol. 74 No. 4, pp. 459-474.
- Bauer, B.C., Johnson, C.D. and Singh, N. (2018), "Place-brand stereotypes: does stereotype-consistent messaging matter?", *Journal of Product and Brand Management*, Vol. 27 No. 7, pp. 754-767.
- Biernat, M. and Kobrynowicz, D. (1997), "Gender-and race-based standards of competence: lower minimum standards but higher ability standards for devalued groups", *Journal of Personality and Social Psychology*, Vol. 72 No. 3, pp. 544-557.
- Bloomfield, I. (2016), "Religious stereotypes", *Contact (online)*, INSERIRE, pp. 11-18.
- Brown, R. (1986), *Social Psychology*, 2nd ed. 2003, Simon & Schuster, Free Press, New York.
- Brown, D., Brown, D. and Anastasopoulos, V. (2002), *Women on Boards: Not Just the Right Thing...but the "Bright" Thing*, The Conference Board of Canada, Ottawa.
- Canadian Human Rights Act, online, available at: <https://www.chrc-ccdp.gc.ca/eng/content/what-discrimination>.
- Carr, D. and Friedman, M. (2005), "Is obesity stigmatizing? Body weight, perceived discrimination, and psychological well-being in the United States", *Journal of Health and Social Behavior*, Vol. 46 No. 3, pp. 244-259.
- Carter, D., Simkins, B. and Simpson, W. (2003), "Corporate governance, board diversity, and firm value", *Financial Review*, Vol. 38 No. 1, pp. 33-53.
- Castagnetti, C. and Rosti, L. (2013), "Unfair tournaments: gender stereotyping and wage discrimination among Italian graduates", *Gender and Society*, Vol. 27 No. 5, pp. 630-658.
- Chan, S.C., Au, A.M. and Lai, S.M. (2020), "The detrimental impacts of negative age stereotypes on the episodic memory of older adults: does social participation moderate the effects?", *BMC Geriatrics*, Vol. 20 No. 1, pp. 1-12.

- Cherry, K. (2020), "How people's prejudices develop", available at: <https://www.verywellmind.com/what-is-prejudice-2795476>.
- Cipolla, C.M. (1962), *The Economic History of World Population*, Penguin Books, Baltimore.
- Collins, P.H. (2015), "Intersectionality's definitional dilemmas", *Annual Review of Sociology*, Vol. 41, pp. 1-20, available at: <https://www.annualreviews.org/doi/pdf/10.1146/annurev-soc-073014-112142>.
- Collins, P. and Bilge, S. (2020), *Intersectionality*, John Wiley & Sons, New York.
- COM, European Commission communication to the European Council and Parliament (2002), available at: http://europa.eu/legislation_summaries/enterprise/industry/n26027_en.htm.
- Comer, L. and Drollinger, T. (1997), "Looking inside the "glass walls": the case of women on the industrial sales force", *Equal Opportunities International*, Vol. 16 No. 4, pp. 1-18.
- Crosby, F.J., Williams, J.C. and Biernat, M. (2004), "The maternal wall", *Journal of Social Issues*, Vol. 60 No. 4, pp. 675-682.
- Diekmann, A. and Eagly, A. (2000), "Stereotypes as dynamic constructs: women and men of the past, present, and future", *Personality and Social Psychology Bulletin*, Vol. 26 No. 10, pp. 1171-1188.
- Dion, K. (2002), "The social psychology of perceived prejudice and discrimination", *Canadian Psychology/Psychologie Canadienne*, Vol. 43 No. 1, p. 1.
- Dovidio, J.F. and Jones, J.M. (2019), "Prejudice, stereotyping, and discrimination", in Baumeister, R.F. and Finkel, E.J. (Eds), *Advanced Social Psychology*, 2nd ed., Oxford University Press, New York, pp. 275-298.
- Eagly, A. and Karau, S. (2002), "Role congruity theory of prejudice toward female leaders", *Psychological Review*, Vol. 109 No. 3, pp. 573-598.
- Eagly, A., Wood, W. and Diekmann, A. (2000), "Social role theory of sex differences and similarities: a current appraisal", in Eckes, T. and Trautner, H. (Eds), *The Developmental Social Psychology of Gender*, Lawrence Erlbaum, Mahwah, NJ, pp. 123-174.
- Eden, D. (1984), "Self-fulfilling prophecy as a management tool: harnessing Pygmalion", *Academy of Management Review*, Vol. 9 No. 1, pp. 64-73.
- Ferber, M. and Nelson, J. (1993), *Feminist Economics Today: beyond Economic Man*, last ed. 2020, University of Chicago Press, Chicago, IL.
- Fiske, S., Cuddy, A., Glick, P. and Xu, J. (2002), "A model of (often mixed) stereotype content: competence and warmth respectively follow from perceived status and competition", *Journal of Personality and Social Psychology*, Vol. 82 No. 6, p. 878.
- Forrester, J.W. (1971), *Counterintuitive Behavior of Social Systems*, Wright-Allen Press, Cambridge, MA, pp. 211-244, Collected papers of J. W. Forrester.
- Gendered Innovations (2020), *Stereotypes*, Stanford University, Stanford, CA, available at: <https://genderedinnovations.stanford.edu/terms/stereotypes.html>.
- Geus (de), A. (1988), "Planning as learning", *Harvard Business Review*, Vol. 66 No. 2, pp. 70-74.
- Gilovich, T., Griffin, D. and Kahneman, D. (Eds) (2002), *Heuristics and Biases: The Psychology of Intuitive Judgment*, Cambridge University Press, Cambridge.
- Green, L. (1998), "Stereotypes: negative racial stereotypes and their effect on attitudes toward African-Americans", *Perspectives on Multiculturalism and Cultural Identity*, Vol. 11 No. 1, pp. 560-573.
- Greenhaus, J. and Parasuraman, S. (1999), "Research on work, family, and gender: current status and future directions", in Powell, G. (Ed.), *Handbook of Gender and Work*, Sage, Thousand Oaks, CA, pp. 391-412.
- Hamilton, D. and Trolie, T. (1986), "Stereotypes and stereotyping: an overview of the cognitive approach", in Dovidio, John F. and Gaertner, Samuel L. (Eds), *Prejudice, Discrimination and Racism*, Academic Press, Orlando and London.
- Heilman, M.E. and Parks-Stamm, E.J. (2007), "Gender stereotypes in the workplace: obstacles to women's career progress", *Social Psychology of Gender*, Emerald Group Publishing, Vol. 4, pp. 47-77.

- Hentschel, T., Heilman, M. and Peus, C. (2019), "The multiple dimensions of gender stereotypes: a current look at men's and women's characterizations of others and themselves", *Frontiers in Psychology*, Vol. 10, p. 11, doi: [10.3389/fpsyg.2019.00011/full](https://doi.org/10.3389/fpsyg.2019.00011/full).
- Hillman, A.J., Shropshire, C. and Canella, A.A. (2007), "Organizational predictors of women on corporate boards", *Academy of Management Journal*, Vol. 50 No. 4, pp. 941-952.
- Holvino, E. (2010), "Intersections: the simultaneity of race, gender and class in organization studies", *Gender, Work and Organization*, Vol. 17 No. 3, pp. 248-277.
- Homan, A., Van Knippenberg, D., Van Kleef, G. and De Dreu, C. (2007), "Bridging faultlines by valuing diversity: diversity beliefs, information elaboration, and performance in diverse work groups", *Journal of Applied Psychology*, Vol. 92 No. 5, p. 1189.
- Huberman, A. (1994), "Data management and analysis methods", in Denzin, N.K. and Lincoln, Y.S. (Eds), *Handbook of Qualitative Research*, Sage Publications, Thousand Oaks, pp. 428-444.
- Hudak, M.A. (1993), "Gender schema theory revisited: men's stereotypes of American women", *Sex Roles*, Vol. 28 Nos 5-6, pp. 279-293.
- Jackson, S.E. and Ruderman, M.N. (Eds) (1995), in *Diversity in Work Teams: Research Paradigms for a Changing Workplace*, American Psychological Association, Washington, DC, pp. 161-186.
- Johnson-Laird, P. (1983), *Mental Models: towards a Cognitive Science of Language, Inference, and Consciousness*, Harvard University Press, Cambridge, MA.
- Kahneman, D. (2011), *Thinking, Fast and Slow*, Farrar, Straus and Giroux, New York.
- Kaschula, R., Maseko, P., Kiesling, Scott F. and Rangel, E.S. (2012), "Intercultural communication and vocational language learning in South Africa: law and healthcare", in Christina Bratt, P. (Ed.), *The Handbook of Intercultural Discourse and Communication*, John Wiley & Sons, New York.
- Klarsfeld, A. (Ed.) (2010), in *International Handbook on Diversity Management at Work: Country Perspectives on Diversity and Equal Treatment*, Edward Elgar, Cheltenham.
- Klump, T. and Su, X. (2013), "A theory of perceived discrimination", *Economic Theory*, Vol. 53 No. 1, pp. 153-180, available at: <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.706.5753&rep=rep1&type=pdf>.
- Koivu, K.L. and Damman, E.K. (2015), "Qualitative variations: the sources of divergent qualitative methodological approaches", *Quality and Quantity*, Vol. 49 No. 6, pp. 2617-2632.
- Konrad, A., Ritchie, J., Jr., Lieb, P. and Corrigan, E. (2000), "Sex differences and similarities in job attribute preferences: a meta-analysis", *Psychological Bulletin*, Vol. 126 No. 4, p. 593.
- Kramer, V.W., Konrad, A.M. and Erkut, S. (2006), *Critical Mass on Corporate Boards: Why Three or More Women Enhance Governance*, Wellesley Centers for Women, Report WCW 11.
- Krishnan, H. and Park, D. (2005), "A few good women: on top management teams", *Journal of Business Research*, Vol. 58, pp. 1712-1720.
- Lazear, E.P. and Rosen, S. (1981), "Rank-order tournaments as optimum labor contracts", *Journal of Political Economy*, Vol. 89 No. 5, pp. 841-864.
- Lazear, E.P. and Rosen, S. (1990), "Male-female wage differentials in job ladders", *Journal of Labor Economics*, Vol. 8 No. 1, Part 2, pp. S106-S123.
- Leonard, D. and Sensiper, S. (1998), "The role of tacit knowledge in group innovation", *California Management Review*, Vol. 40 No. 3, pp. 112-132.
- Lippmann, W. (1922), "Public opinion", available at: <https://books.google.it/books?id=fqpEWJHEzIkC&printsec=frontcover&dq=Lippmann+public+opinion&hl=en&sa=X&ved=2ahUKEwj13YHA9KHqAhVHmYsKHVlzD8oQ6AEwAHoECAAAQAg#v=onepage&q=Lippmann%20public%20opinion&f=false>.
- Locke, L. and Richman, C. (1999), "Attitudes toward domestic violence: race and gender issues", *Sex Roles*, Vol. 40 No. 3, pp. 227-247.

- Mahoney, J. (2007), "Qualitative methodology and comparative politics", *Comparative Political Studies*, Vol. 40 No. 2, pp. 122-144, available at: <http://www.comunicacaoeleitoral.ufpr.br/wp-content/uploads/2019/08/Mahoney-Qualitative-Methodology.pdf>.
- Marx, D.M. and Ko, S.J. (2012), "Prejudice, discrimination, and stereotypes (racial bias)", in *Encyclopedia of Human Behavior*, 2nd ed., available at: <https://www.sciencedirect.com/referencework/9780080961804/encyclopedia-of-human-behavior>.
- McKinsey and Company (2007), *Women Matter. Gender Diversity, a Corporate Performance Driver*, McKinsey & Company, Paris, available at: http://www.mckinsey.com/locations/swiss/news_publications/pdf/women_matter_english.pdf.
- Mella, P. (2008), "Systems thinking: the art of understanding the dynamics of systems", *International Journal of Learning*, Vol. 15 No. 10, pp. 79-88.
- Mella, P. (2012), *Systems Thinking: Intelligence in Action*, Springer, New York, NY.
- Mella, P. (2014), *The Magic Ring*, 2nd ed., Springer, New York, NY, p. 2021.
- Mella, P. (2017), *The Combinatory Systems Theory*, Springer, New York, NY.
- Mella, P. (2018), "The law of increasing productivity", *International Journal of Markets and Business Systems (IJMABS)*, Vol. 3 No. 4, pp. 297-316.
- Mella, P. (2020), *Constructing Reality: the "Operationalization" of Bateson's Conjecture on Cognition*, Springer Briefs in Cognition, Springer, Cham.
- Mella, P. and Colombo, C. (2012), "The wheels of change in organizations", *International Journal of Knowledge, Culture and Change Management*, Vol. 11 No. 6, pp. 247-265.
- Movahedi, S. (1978), "The drug addict and addiction: cultural stereotypes and clinical theories", *Urban Life*, Vol. 7 No. 1, pp. 45-66.
- Nelson, T. (Ed.) (2009), *Handbook of Prejudice, Stereotyping, and Discrimination*, Psychology Press, Taylor&Francis Group, New York.
- Nelson, R. and Winter, S. (1982), *An Evolutionary Theory of Economic Change*, Belknap Press of Harvard University, Cambridge, MA.
- O", D. (2012), "Thinking, fast and slow by Daniel Kahneman (book review)", *Journal of Social, Evolutionary, and Cultural Psychology*, Vol. 6 No. 2, pp. 253-256.
- Pascoe, E.A. and Smart Richman, L. (2009), "Perceived discrimination and health: a meta-analytic review", *Psychological Bulletin*, Vol. 135 No. 4, p. 531, available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2747726/>.
- Paterson, B., Dubouloz, C.J., Chevrier, J., Ashe, B., King, J. and Moldoveanu, M. (2009), "Conducting qualitative metasynthesis research: insights from a metasynthesis project", *International Journal of Qualitative Methods*, Vol. 8 No. 3, pp. 22-33, available at: <https://journals.library.ualberta.ca/ijqm/index.php/IJQM/article/view/5100>.
- Plous, S. and Williams, T. (1995), "Racial stereotypes from the days of American slavery: a continuing legacy 1", *Journal of Applied Social Psychology*, Vol. 25 No. 9, pp. 795-817.
- Polanyi, M. (2015), *Personal Knowledge: Towards a Post-critical Philosophy*, 1st ed. 1990, The University of Chicago Press, Chicago, IL.
- Powell, G. and Butterfield, D. (1994), "Investigating the "glass ceiling" phenomenon: an empirical study of actual promotions to top management", *Academy of Management Journal*, Vol. 37, pp. 68-86.
- Powell, G., Butterfield, D. and Parent, J. (2002), "Gender and managerial stereotypes: have the times changed?", *Journal of Management*, Vol. 28 No. 2, pp. 177-193.
- Reynolds, D. (2007), "Restraining Golem and harnessing Pygmalion in the classroom: a laboratory study of managerial expectations and task design", *Academy of Management Learning and Education*, Vol. 6 No. 4, pp. 475-483.
- Roberson, L. and Kulik, C. (2007), "Stereotype Threat at work", *Academy of Management Perspectives*, Vol. 21 No. 2, pp. 24-40.

- Rosenthal, R. and Jacobsen, L. (1968), *Pygmalion in the Classroom*, Rinehart and Winston, New York.
- Rosenthal, L. and Overstreet, N. (2015), "Stereotyping", *Encyclopedia of Mental Health*, 2016, 2nd ed. ed., pp. 225-229.
- Schein, E.H. (1990), "Organizational culture", *American Psychological Association*, Vol. 45 No. 2, p. 109.
- Schein, V., Mueller, R., Lituchy, T. and Liu, J. (1996), "Think manager—think male: a global phenomenon?", *Journal of Organizational Behavior*, Vol. 17 No. 1, pp. 33-41.
- Schiebinger, L. (2014), "Gendered innovations: harnessing the creative power of sex and gender analysis to discover new ideas and develop new technologies", *Triple Helix*, Vol. 1 No. 1, pp. 1-17.
- Scott, K.A., Heathcote, J.M. and Gruman, J.A. (2011), "The diverse organization: finding gold at the end of the rainbow", *Human Resource Management*, Vol. 50 No. 6, pp. 735-755.
- Senge, P. (1990), *The Fifth Discipline: the Art and Practice of the Learning Organization*, 1st ed., Doubleday/Currency, New York, NY.
- Senge, P. (2006), *The Fifth Discipline: the Art and Practice of the Learning Organization*, Doubleday/Currency, New York, NY, (last edition, revised and enlarged).
- Singh, V. and Vinnicombe, S. (2004), "Why so few women directors in top UK boardrooms? Evidence and theoretical explanations", *Corporate Governance: An International Review*, Vol. 12 No. 4, pp. 479-488.
- Smith, A. (1776), *An Inquiry into the Nature and Causes of the Wealth of Nations*, Methuen, London, available at: http://www.gutenberg.org/catalog/world/readfile?fk_files=3275283.
- Smith, E.R., Miller, D.A., Maitner, A.T., Crump, S.A., Garcia- Marques, T. and Mackie, D.M. (2006), "Familiarity can increase stereotyping", *Journal of Experimental Social Psychology*, Vol. 42, pp. 471-478.
- SPEEXX on line, available at: <https://www.speexx.com/it/speexx-blog/stereotipi-italiani/>.
- Spender, J.C. (1993), "Competitive advantage from tacit knowledge? Unpacking the concept and its strategic implications", *Academy of Management Proceedings*, Briarcliff Manor, Vol. 1993 No. 1, pp. 37-41.
- Steele, C. (1997), "A Threat in the air: how stereotypes shape intellectual identity and performance", *American Psychologist*, Vol. 52 No. 6, pp. 613-629.
- Sterman, J.D. (2000), *Business Dynamics: Systems Thinking and Modeling for a Complex World*, McGraw-Hill, New York.
- Thomas-Hunt, M. and Phillips, K. (2004), "When what you know is not enough: expertise and gender dynamics in task groups", *Personality and Social Psychology Bulletin*, Vol. 30 No. 12, pp. 1585-1598.
- UNESCO (2016), "Incheon Declaration and Framework for Action for the implementation of Sustainable Development Goal 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all", available at: <https://www.un.org/youthenvoy/2020/12/incheon-declaration-and-framework-for-action-for-the-implementation-of-sustainable-development-goal-4/>.
- Valian, V. (1998), *Why So Slow? the Advancement of Women*, MIT Press, Cambridge, 9 – Heiman.
- WGEPAD - Working Group of Experts on People of African Descent (General Assembly, 74th Session (2019), "Elimination of racism, racial discrimination, xenophobia and related intolerance: comprehensive implementation of and follow-up to the Durban Declaration and Programme of Action", available at: <https://documents-dds-ny.un.org/doc/UNDOC/GEN/N19/243/71/PDF/N1924371.pdf?OpenElement>.
- Wikipedia.com (2020), "Stereotype", available at: <https://en.wikipedia.org/wiki/Stereotype>.
- Williams, J. (2004), "The maternal wall", *Harvard Business Review*, Vol. 82 No. 10, pp. 26-27.

Williams, D.R., Yu, Y., Jackson, J.S. and Anderson, N.B. (1997), "Racial differences in physical and mental health: socio-economic status, stress and discrimination", *Journal of Health Psychology*, Vol. 2 No. 3, pp. 335-351.

Zawisza, M. (2018), "The terrifying power of stereotypes – and how to deal with them", available at: <https://theconversation.com/the-terrifying-power-of-stereotypes-and-how-to-deal-with-them-101904>.

Further reading

Carter, D., D, F., Simkins, B. and Simpson, W. (2010), "The gender and ethnic diversity of US boards and board committees and firm financial performance", *Corporate Governance: An International Review*, Vol. 18 No. 5, pp. 396-414.

Malmström, M., Johansson, J. and Wincent, J. (2017), "Gender stereotypes and venture support decisions: how governmental venture capitalists socially construct entrepreneurs' potential", *Entrepreneurship Theory and Practice*, Vol. 41 No. 5, pp. 833-860.

Morrison, A., White, R., White, R. and Van Velsor, E. (1987), *Breaking the Glass Ceiling: Can Women Reach the Top of America's Largest Corporations?*, Pearson Education, New York.

Newman, T. (2021), "Sex and gender: what is the difference?", available at: <https://www.medicalnewstoday.com/articles/232363>.

Richmond, B. (1994), "System dynamics/systems thinking: let's just get on with it", in *Sterling: International Systems Dynamics Conference*, available at: http://webspace.webring.com/people/ah/himadri_banerji/pdf/systhnk.pdf.

Rosen, S. (1981), *The Economics of Superstars*, American Economic Review, p. 71.

Rosenthal, R. (2002), "The Pygmalion effect and its mediating mechanisms", in Aronson, J. (Ed.), *Improving Academic Achievement: Impact of Psychological Factors on Education*, Academic Press, Elsevier, Cambridge, MA, pp. 25-36.

Stiglitz, J.E. (1975), "The theory of "Screening, Education, and the Distribution of Income"", *The American Economic Review*, Vol. 3, p. 283300.

UN (2020), "Agenda 2030. SDG 5: achieve gender equality and empower all women and girls", available at: <https://www.unwomen.org/en/news/in-focus/women-and-the-sdgs/sdg-5-gender-equality>.

Corresponding author

Piero Mella can be contacted at: piero.mella@unipv.it

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